PROPOSAL

November 1, 2026

1 Idea & How I got here

I find it interesting to examine how our brains actually think, and how much task-dependent structure there is. Much of this idea has been investigated [1] in some depth, with a dynamical mean-field theory (DMFT) for recurrent neural networks, with a parameter γ that determines whether the network is being trained in the reservoir regime ($\gamma \to 0^+$) or in a regime where weights are structured by correlations in the expected task outputs (to a degree specified by the magnitude of γ). Here, the activation functions $\phi(\cdot)$ used are nonlinear $\tanh$ and the system considers only one-to-one interactions.

However, there may be value (as the authors mention in SI7) in considering a p-neuron generalisation. When ϕ is taken to be linear (e.g. $\phi(x) = x$), the DMFT equations become much simpler and may be solved (in principle) without the expensive Monte Carlo processes used by [1]. Further, when $p > 1$, the linear activations can generate nonlinear and chaotic dynamics, and deeper analysis of this model has not been done in the regime where $\gamma > 0$.

There is some biological motivation to this project [2]. However, a problem is to consider how to bound the linear activations, and it has been suggested to constrain activations to a fixed spherical norm [1]. If there turn out to be ways to investigate if other constraints work without sacrificing the solvability of the DMFT, that will also be interesting. Further, the single-neuron statistics remain Gaussian (due to the linear activation), and are hence more tractable.

Therefore the question I hope to ask will be: "Can a linear p-neuron network be trained to exhibit responses similar to those of the $p = 1$ nonlinear case? What is the theory that governs its behavior, and can the DMFT be made solvable?"

I will start by writing a model with a tunable p, and try out a toy problem (like the sine wave) before attempting to replicate the macaque results. I will also attempt to solve the DMFT in whichever way makes sense (I will probably need some advice on this!).

References

[1] David G. Clark, Blake Bordelon, Jacob A. Zavatone-Veth, and Cengiz Pehlevan. Structure, disorder, and dynamics in task-trained recurrent neural circuits, May 2026.

[2] Leo Kozachkov, Jean-Jacques Slotine, and Dmitry Krotov. Neuron–astrocyte associative memory. *Proceedings of the National Academy of Sciences*, 122(21):e2417788122, May 2025.